

Graph AI for Broadening the National Tax Net

Theme: FinTech, Knowledge Graphs, Fraud Detection

The Core Problem:

Pakistan struggles with a historically low tax-to-GDP ratio, driven by a massive undocumented economy. While the Federal Board of Revenue (FBR) and excise departments hold fragmented databases (vehicle registration, real estate transactions, luxury travel logs, utility bills), these data siloes are rarely connected intelligently to identify high-net-worth individuals who are non-filers or under-reporting income.

The Challenge:

Using synthetic or anonymized localized datasets mimicking Pakistani civic data, teams must build a Knowledge Graph and Entity Resolution pipeline. The AI must link disparate identities (e.g., matching a person who owns a 2000cc vehicle in the Excise database, has a 300,000 PKR monthly electricity bill, but files a zero-tax return). The system must output a "Tax Compliance Deviation Score" and automatically generate an audit trail explaining why this individual was flagged.

- **Academic Complexity:** Unsupervised entity resolution, graph neural networks (GNNs) for anomaly detection, and handling unstructured Urdu/English mixed text in name/address fields.
- **Industry Applicability:** Enterprise-grade fraud detection for the banking sector (KYC/AML compliance) and sovereign-level GovTech for tax infrastructure.
- **Evaluation Criteria:** Precision and recall of the entity resolution algorithm, the depth of the graph-based relationship mapping, and the interpretability of the AI's audit trail (Explainable AI).